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Experiment: 10 5G Network Slicing Environment Using MATLAB

Aim

To analyze the performance of a 5G Network Slicing environment by studying the number of

network slices, slice utilization, and end-to-end latency using MATLAB.

Objectives

Objective – 1: Analyze the Number of Network Slices

Analyze the creation and allocation of multiple network slices in a 5G network using MATLAB.

Observe how different slices can be configured to support diverse service requirements and

applications.

Objective – 2: Compare Slice Utilization (%)

Compare the utilization of different network slices using MATLAB. Analyze how efficiently

network resources are allocated and utilized across multiple slices.

Objective – 3: Analyze End-to-End Latency (ms)

Analyze the end-to-end latency of different network slices using MATLAB.

Compare the latency

values to understand the Quality of Service (QoS) requirements of various 5G applications.

Objective1:

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
slices = {'eMBB','URLLC','mMTC','Private','IoT'};
```

```
count = [1 1 1 1 1]; % One instance of each network slice
```

```
figure
```

bar(count)

grid on

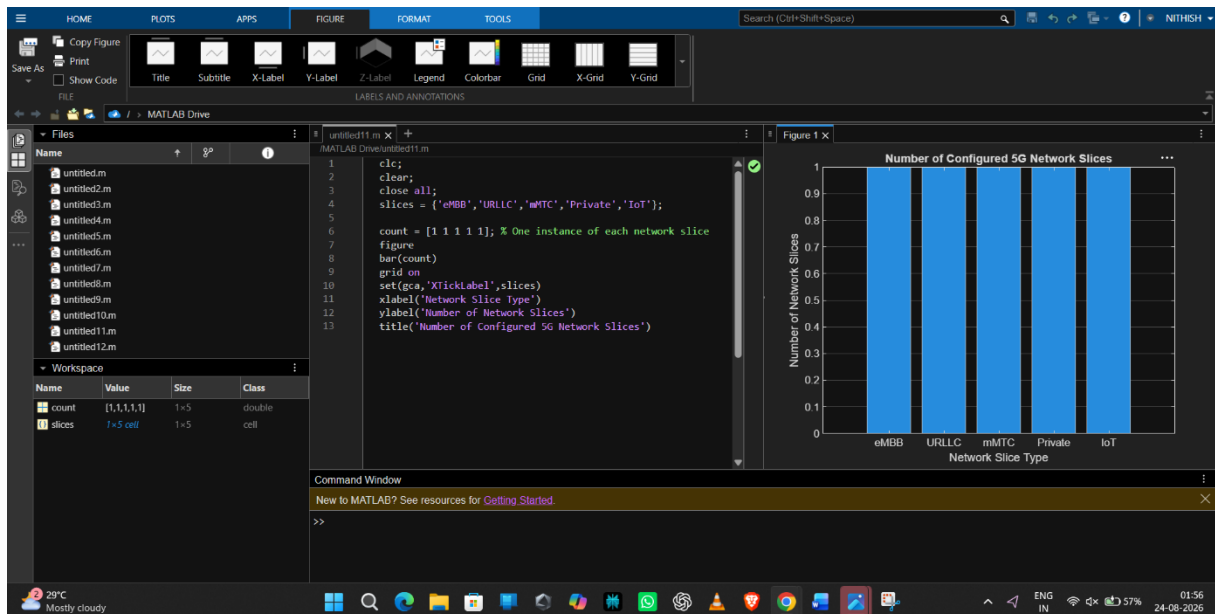
set(gca,'XTickLabel',slices)

xlabel('Network Slice Type')

ylabel('Number of Network Slices')

title('Number of Configured 5G Network Slices')

output:



Objective2:

Program:

clc;

clear;

close all;

slices = categorical({'eMBB','URLLC','mMTC','Private','IoT'});

utilization = [85 70 60 75 50];

figure;

bar(slices, utilization);

grid on;

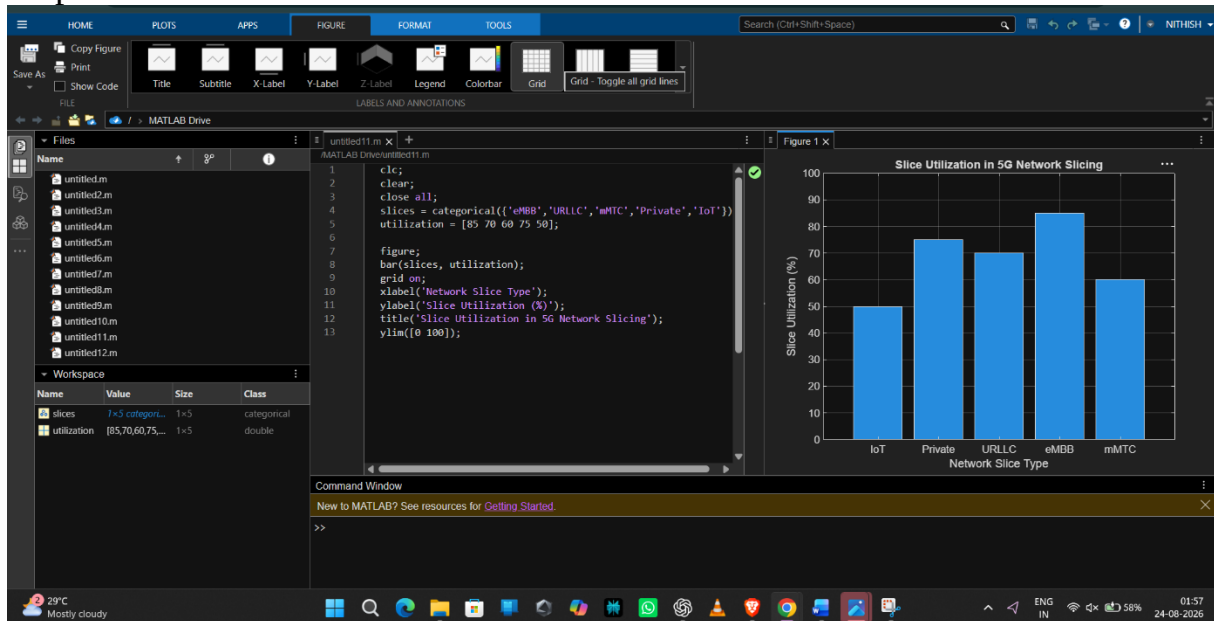
xlabel('Network Slice Type');

ylabel('Slice Utilization (%)');

title('Slice Utilization in 5G Network Slicing');

ylim([0 100]);

output:



Objective3:

Program:

clc;

clear;

close all;

slices = [1 2 3 4 5];

latency = [25 5 40 15 60];

figure;

plot(slices, latency, '-o','LineWidth',2,'MarkerSize',8);

grid on;

xticks(slices);

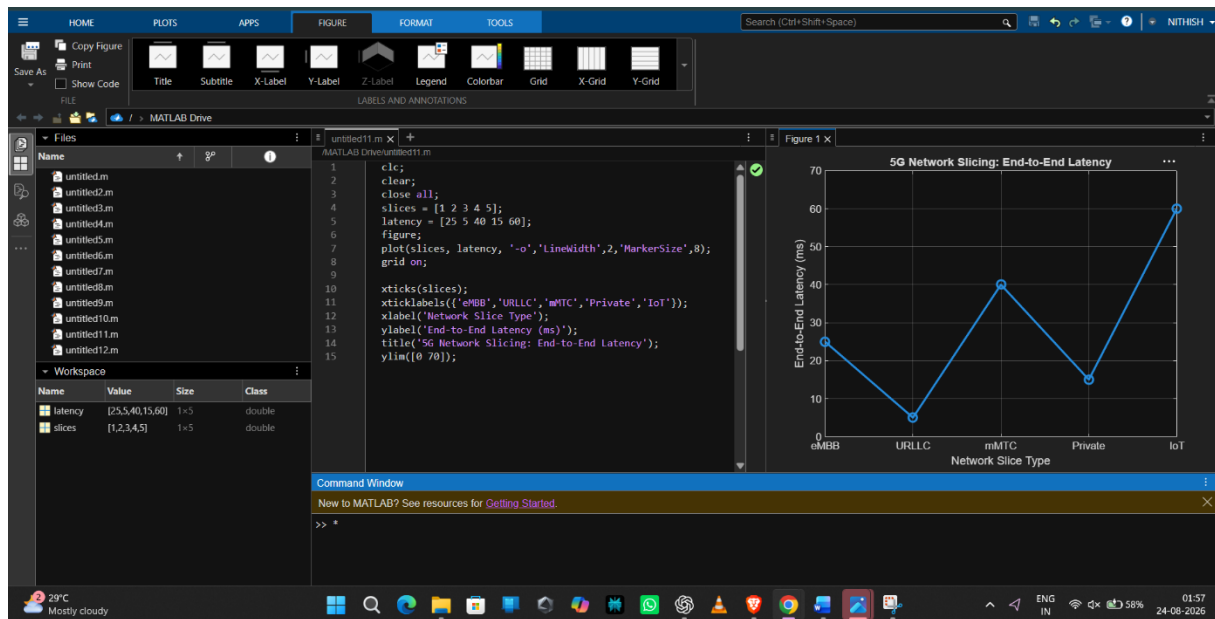
```
xticklabels({'eMBB','URLLC','mMTC','Private','IoT'});
```

```
xlabel('Network Slice Type');
```

```
ylabel('End-to-End Latency (ms)');
```

```
title('5G Network Slicing: End-to-End Latency');
```

output:



Result

The MATLAB analysis successfully demonstrated the characteristics of a 5G Network Slicing environment. The results showed that multiple network slices can coexist while exhibiting different utilization levels and latency characteristics based on their intended services.